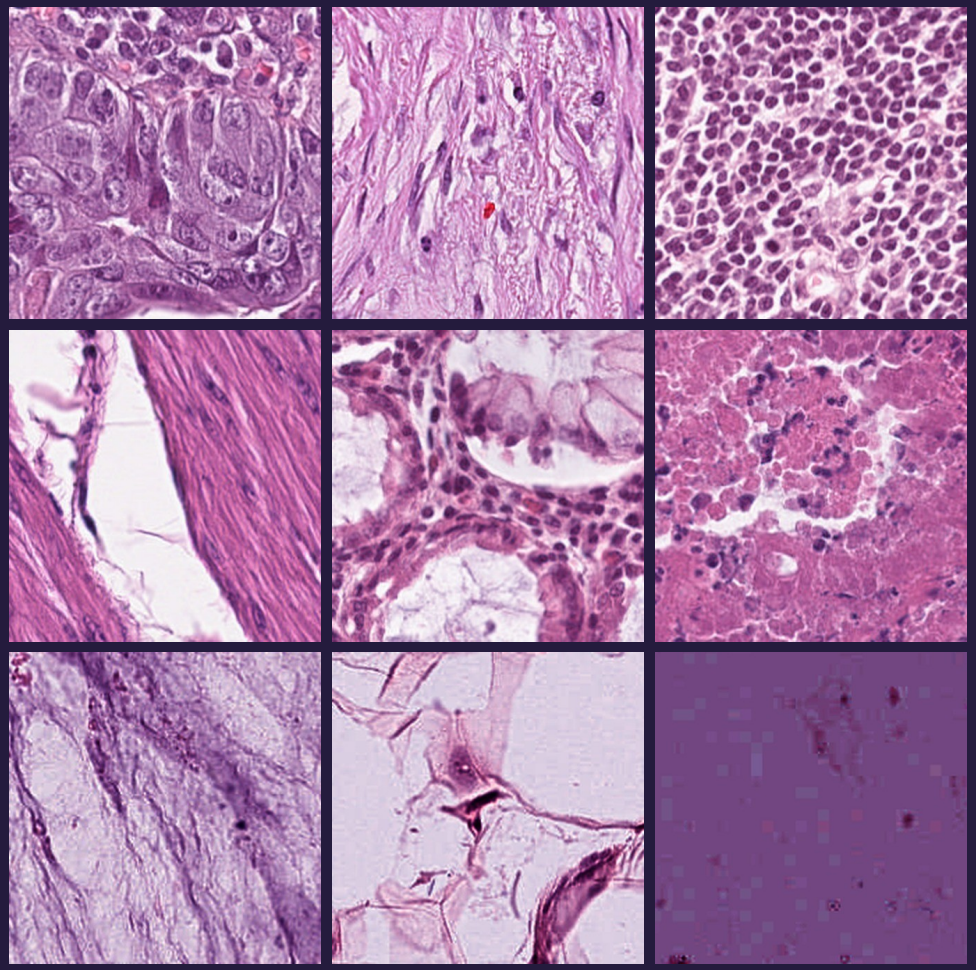
TwinRead

Yubo Wu, Noah Zhang, William Liu, Franklin Liu, Andy Chen

University of Illinois Urbana-Champaign

2026 Ashby Prize in Computational Science Hackathon



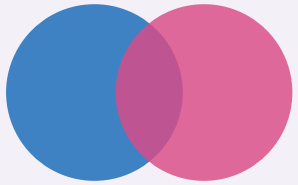
Why a second opinion matters

75.3%

of individual breast-biopsy reads matched the expert panel, and only 48% for atypia

47%

of the world’s population has little or no access to diagnostics



Can two AI models act as independent readers?

And can we make it run offline on a small device?

CONCH and MUSK make different mistakes

CONCH
Contrastive: 1.17 M image–caption pairs

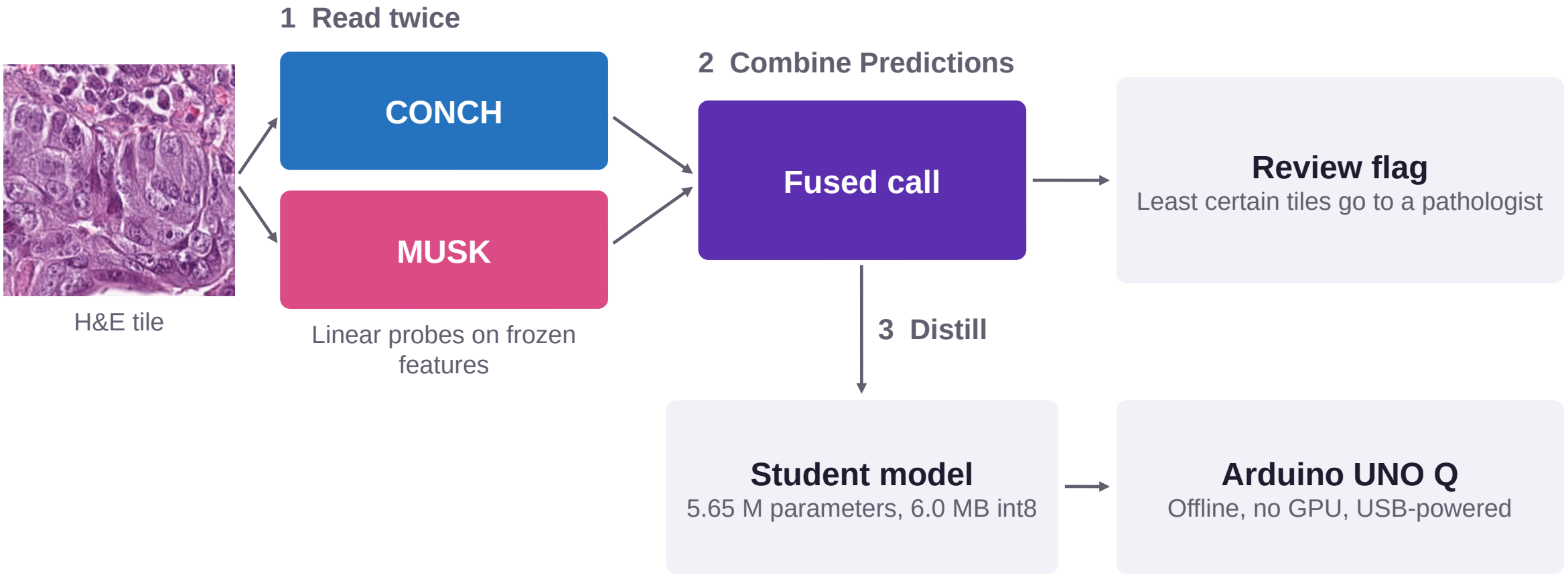
MUSK
Masked: 50 M images and 1 B text tokens

At least one model was wrong on 505 of 7,180 test images



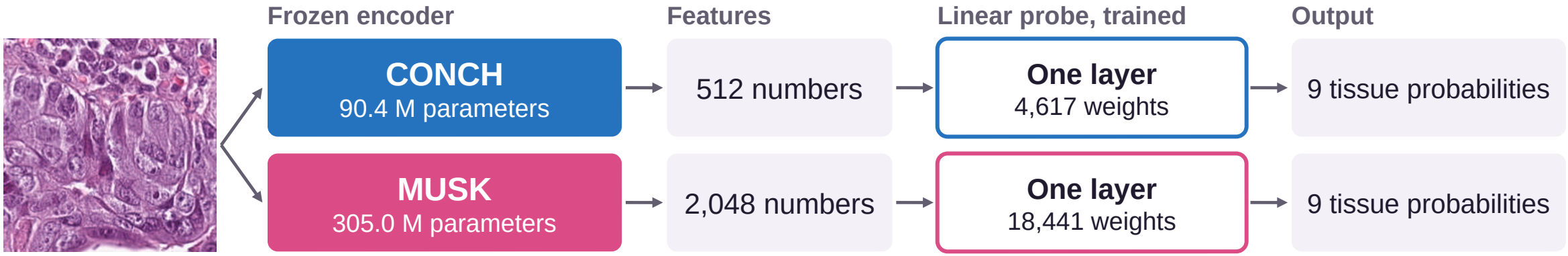
Only 110 out of the 505 were misclassified by both models.

System Architecture



Train: 17,349 NCT-CRC-HE-100K tiles. Test: 7,180 CRC-VAL-HE-7K tiles (9 classes) and 1,500 Camelyon17 tiles from an unseen hospital.

Linear probes: one simple classifier on each model

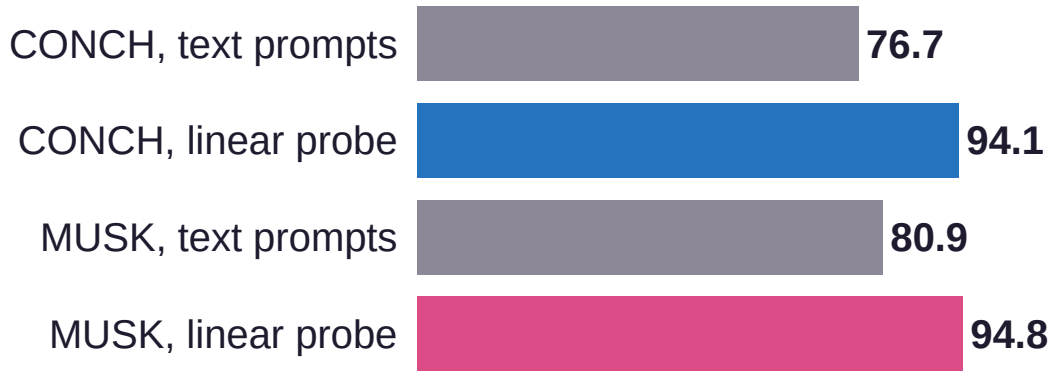


Why a linear probe

- **Honest:** simple enough that it can only use what the features already know
- **Independent:** each probe sees one model, so their errors stay different
- **Cheap:** 17,349 labelled tiles, fitted on CPU; the big models never change

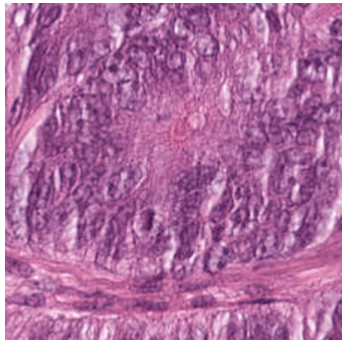
Probes add 14–17 points over text prompts

Balanced accuracy on the colorectal test set (%)



Probe: standardised logistic regression, strength chosen on 2,658 validation tiles (CONCH C = 1, MUSK C = 0.1). Text prompts: zero-shot, no training. Parameters count the vision towers.

Fusion: average the two models' probabilities



Real test tile. Truth: tumour.

CONCH

Tumour 100%

Mucus 0%

Calls tumour

+

MUSK

Tumour 17%

Mucus 83%

Calls mucus

=

Average

Tumour 59%

Mucus 41%

Calls tumour, correctly, but only 59% sure, so the review flag fires

Three rules, chosen on validation only

Rule	Validation correct, of 2,658	Test points vs MUSK
Average (chosen)	**2,641**	**+0.74**
Gated neural network	2,640	−0.40
Logistic regression on joined features	2,639	+0.71

286 → 239

errors, compared with MUSK alone: fusion fixes 154 and adds 107.

Tile chosen by rule: the median, by fused top probability, of the tumour tiles that one model misses and fusion gets right. Test: CRC-VAL-HE-7K.

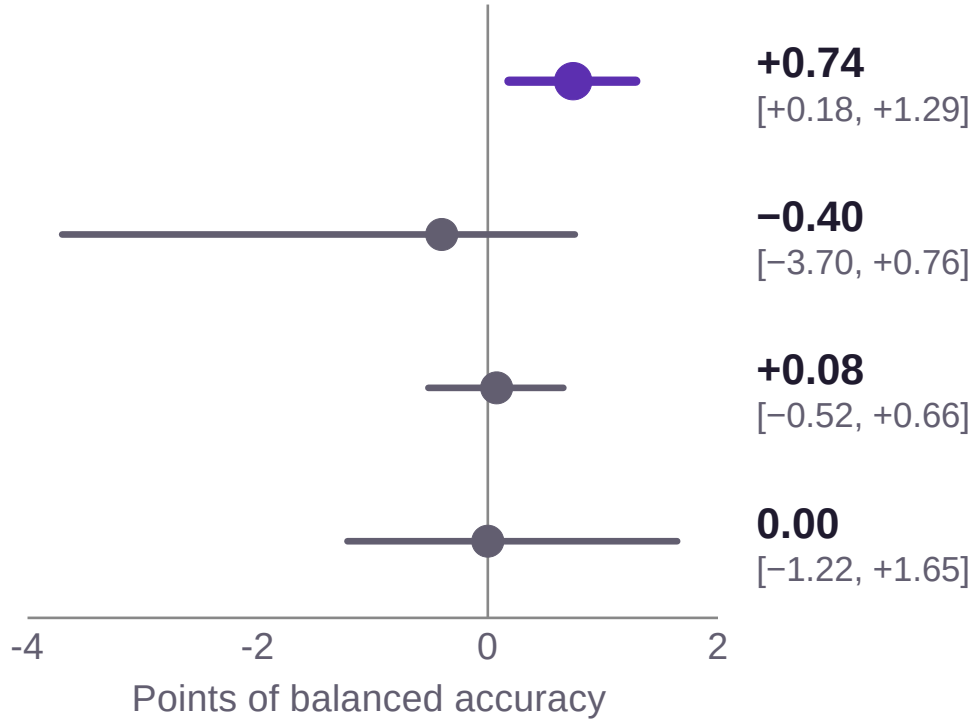
Fusion adds +0.74 points on colorectal tiles only

Balanced accuracy, colorectal test set

- **94.13%** CONCH alone
- **94.80%** MUSK alone
- **95.55%** CONCH + MUSK fused

+3.6 points of tumour recall: most of the net gain

Fused minus best single reader, 95% CI



CIs: tile-level on colorectal (anti-conservative, no slide IDs), slide-clustered on Camelyon17. Test sets: 7,180 and 1,500 tiles.

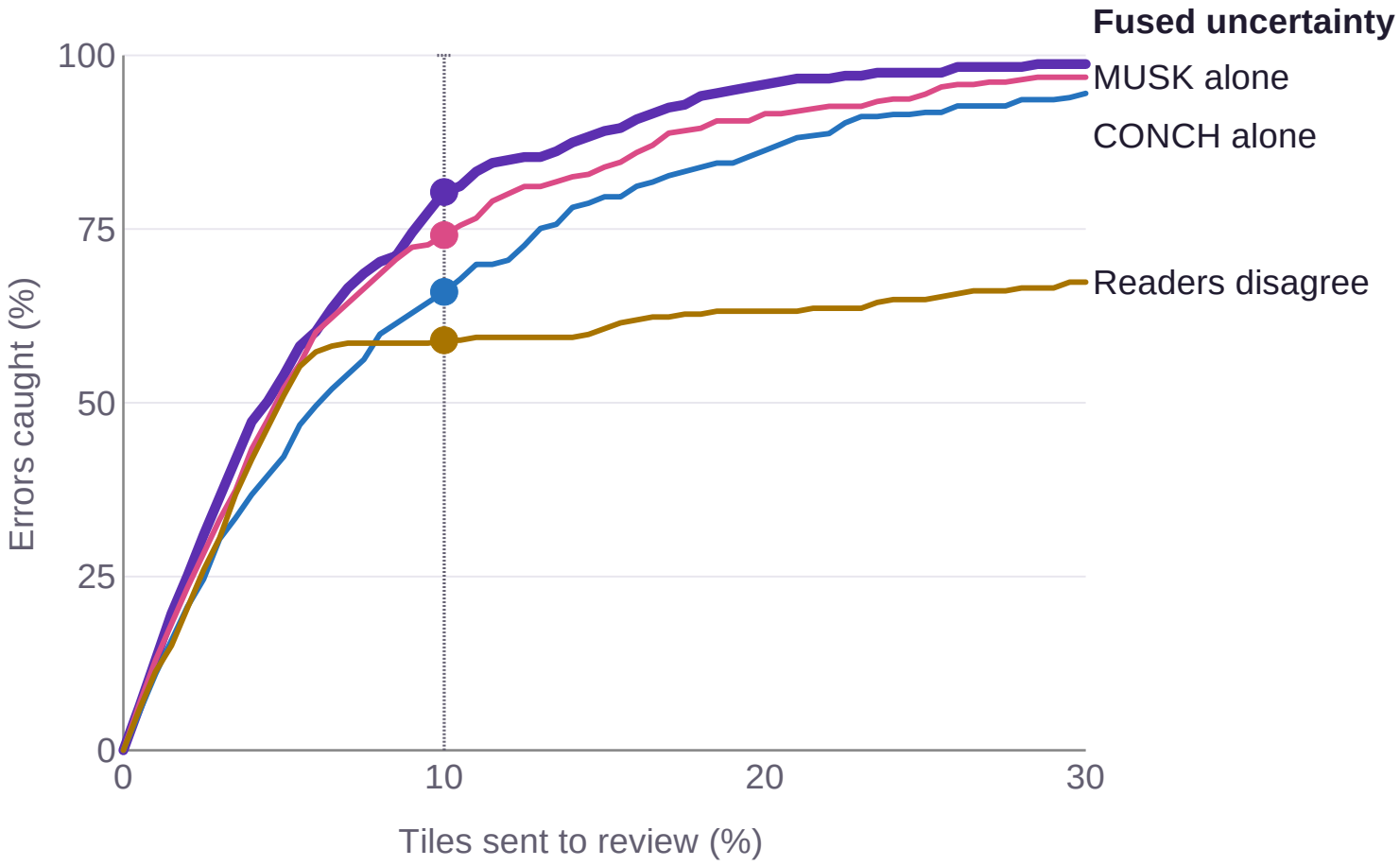
Reviewing the least certain 10% catches 80% of errors

Errors caught at a 10% review budget

- **80.3%**
  Fused reader’s uncertainty

- **74.1%** MUSK alone
- **66.0%** CONCH alone
- **59.0%** Readers disagree

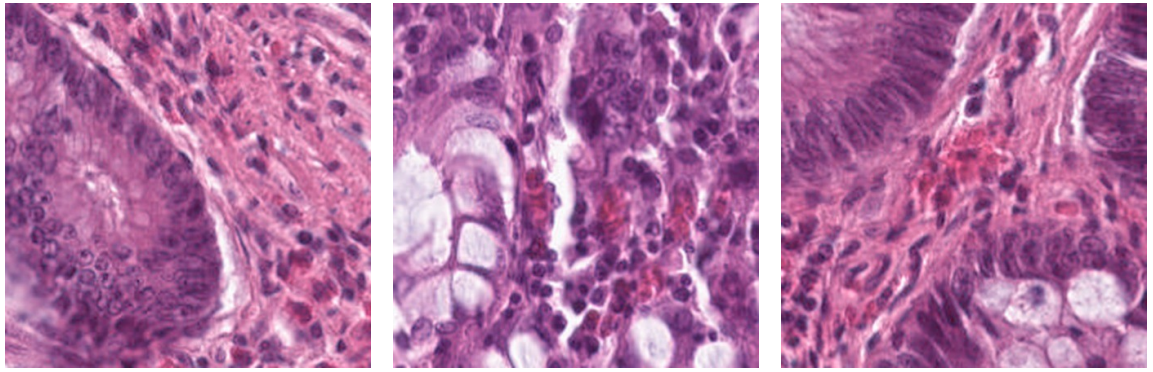
The signal is the fused reader’s doubt, not disagreement.



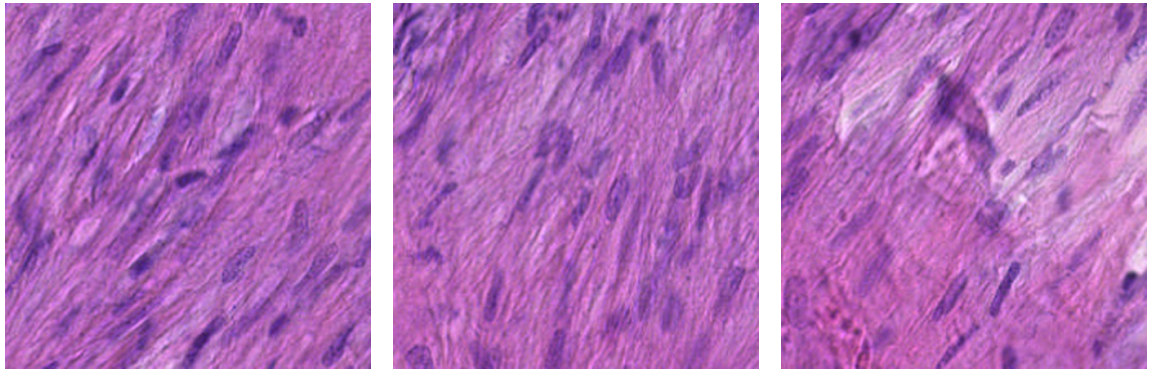
CRC-VAL-HE-7K, 7,180 tiles. Single readers flag their own errors by their own confidence.

53 tiles fooled both readers; six also escaped the flag

Truth: tumour. Both readers: normal.



Truth: stroma. Both readers: muscle.



53 tiles (0.74%): both wrong (>80% confidence)

47 caught by the pre-registered flag, which reviews 25.1% of tiles

6 escaped it (shown): both readers at least 0.98 confident

Two models agreeing is not proof. A pathologist stays in the loop.

CRC-VAL-HE-7K, truth as labelled in the dataset. 36 of the 53 are muscle vs stroma; 11 are tumour called normal.

Why shrink it: AI that works without a network



Why quantise to int8

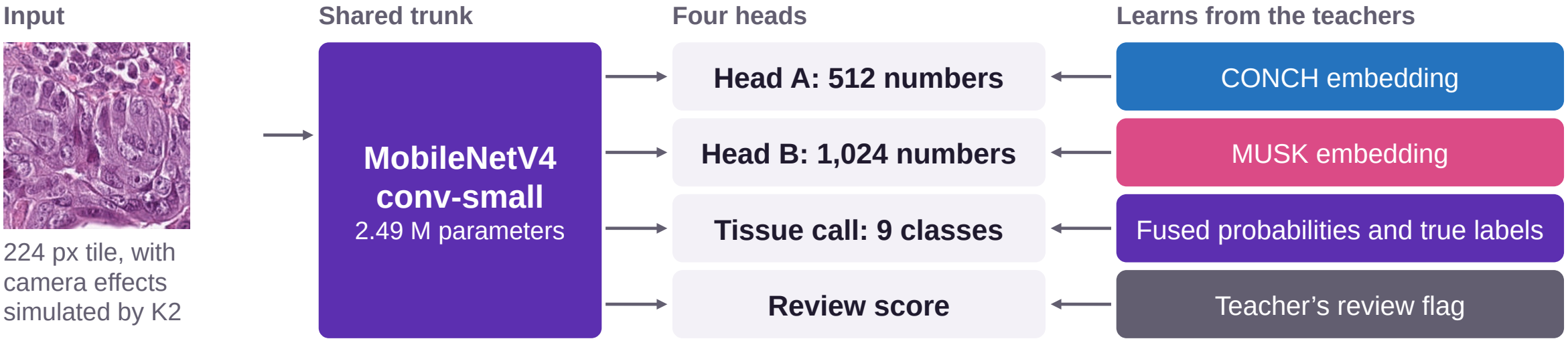
- **Smaller:** 22.6 MB down to 6.0 MB, ×3.8
- **Faster and lighter:** ×1.5 faster and ×1.8 less RAM on x86
- **Small cost:** 94.15% to 93.05% balanced accuracy; 96.8% of calls unchanged

Why run on a small device

- **Low-access clinics:** no GPU, server or network
- **Disaster relief:** Hurricane Maria knocked out 95.6% of Puerto Rico’s cell sites
- **Mars:** up to 22 minutes each way to Earth, so decisions must be local
- **Privacy and cost:** images stay local; $79 list price

Checkpoints include text encoders. Speed and RAM: MobileNetV4 backbone, 4 threads, x86; fp32 not timed on the board. Sources: Wilson et al., Lancet 2018; FCC 2017; McBrayer et al., NASA, 2022; Arduino 2026.

Distillation: one small student learns from both models



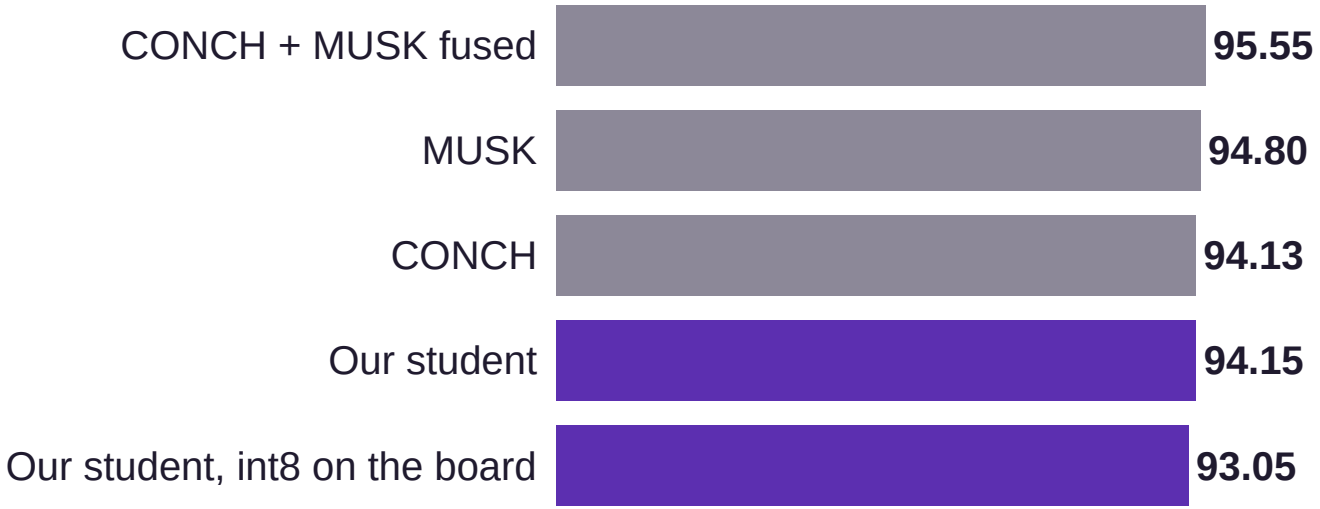
How it trains

- **Copy the teachers:** match each teacher’s embedding, the fused soft labels and the review flag
- **Learn the truth:** true tissue labels as well, so it is not only imitating
- **Survive a camera:** K2 simulates a USB camera filming a screen during training
- **Run anywhere:** teachers are needed for training only, never on the board

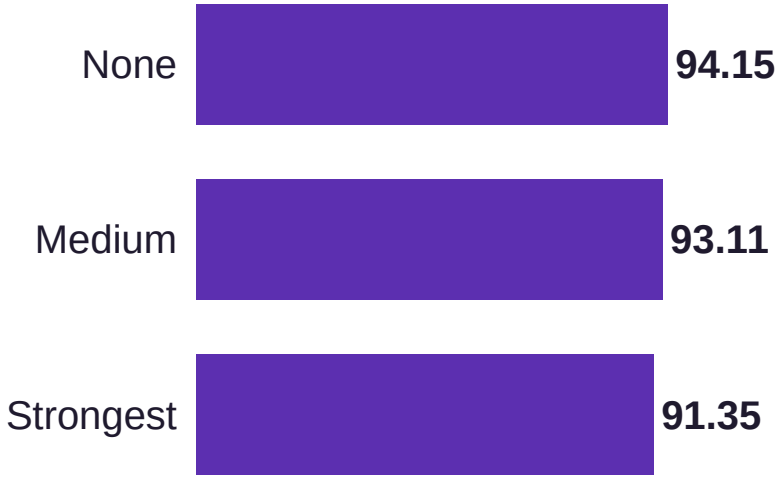
Loss weights: embeddings 1, fused soft labels 1 (T = 2), true labels 0.5, review 0.5. 15 epochs, batch 128. Training targets are out-of-fold. Heads A and B share one trunk, so they are far less independent than CONCH and MUSK.

Our trained student: 1.4% of the size, 2.5 points behind

Balanced accuracy, colorectal test set (%)



Our student under simulated camera effects (%)



15 epochs

on 17,349 labelled tiles, on one RTX 5060

3,880/s

images through a full training step, with K2 camera effects

96.2%

of calls on clean test tiles match the fused teacher

The student is fine-tuned with labels, so it is not a like-for-like test of features. Camera effects are the K2 simulation, not real captures; the teachers were not run under them.

CUDA kernels: ×202 preprocessing, ×10.6 augmentation

×202

K1: dual-reader preprocessing

One tile in, CONCH 448 px and MUSK 384 px out, bit-exact to the official transforms

×10.6

K2: training augmentation

Nine dependent stages plus JPEG, each within one grey level of a float64 reference





×1.11 whole embedding run: the model forward is 89% of the time

×2.49 whole student training step: 1,559 to 3,880 images/s

RTX 5060; compile-checked for Delta A100 and DeltaAI GH200; compute-sanitizer clean. Tiles/s (K1), images/s (K2). K2 simulates a USB camera filming a screen.

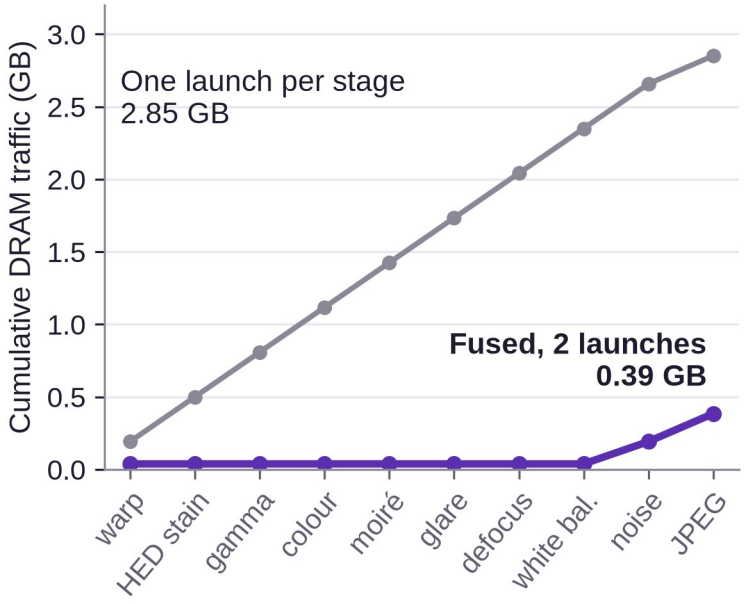
Fusing dependent stages avoids DRAM round trips

A dependent kernel chains stages where each one needs the previous stage's output. **Fusing** runs the chain in one launch, with intermediates in on-chip shared memory.

Why fuse

One launch per stage sends every intermediate through DRAM: ×7.4 the traffic.

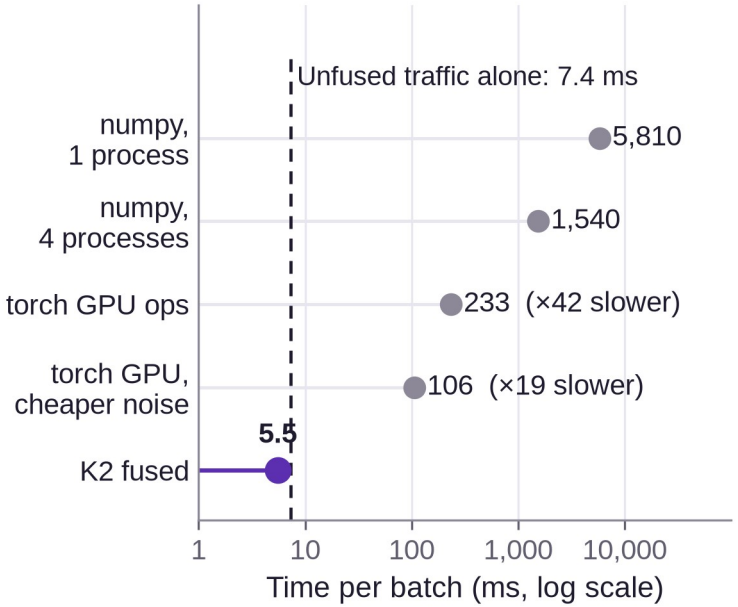
K2, batch 256, image data only, calculated



What it accomplishes

K2 takes 5.5 ms per batch, less than the unfused chain's memory traffic alone.

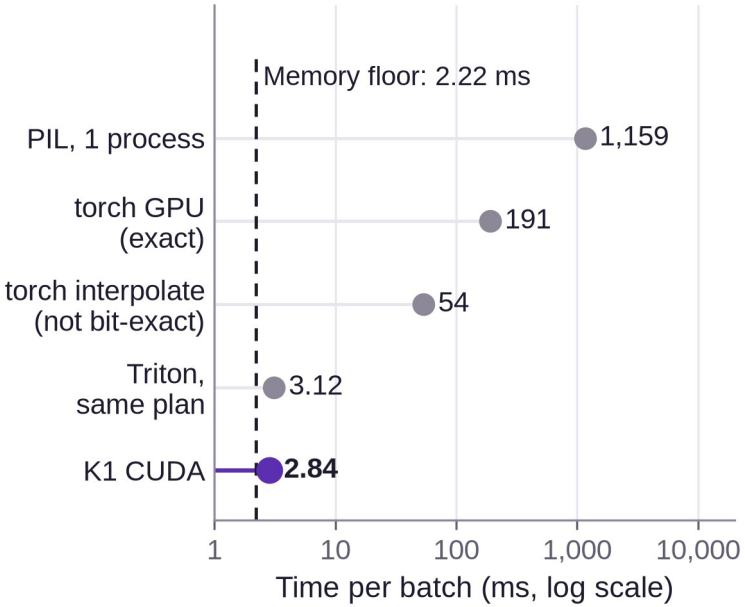
K2, batch 256, severity 1.0, measured



Why it's cool

K1 runs within 28% of the memory-bandwidth floor and stays bit-exact.

K1, batch 256, both readers, measured



DRAM traffic and floors are calculated from tensor sizes and the measured memcpy rate (388 GB/s), not hardware counters. Times are CUDA events on a shared RTX 5060. No hand-written unfused CUDA version was measured.

UNO Q setup: what runs where

Linux side Qualcomm QRB2210, 4× Cortex-A53 at 2.0 GHz, 3.6 GiB RAM, Debian 13, no GPU

Frame in	Crop, tissue check	int8 student	Probes, review rule	Web page
stored tile; USB camera next	numpy	onnxruntime on CPU	numpy	call and flag, local only
5.6 ms	**23.5 ms**	**18.7 ms**	**1.1 ms**	**23.5 ms**

Microcontroller side
STM32U585, Zephyr

Bridge call	LED sketch	8×13 LED matrix
8.2 ms per message	flashed over USB-C	shows the status

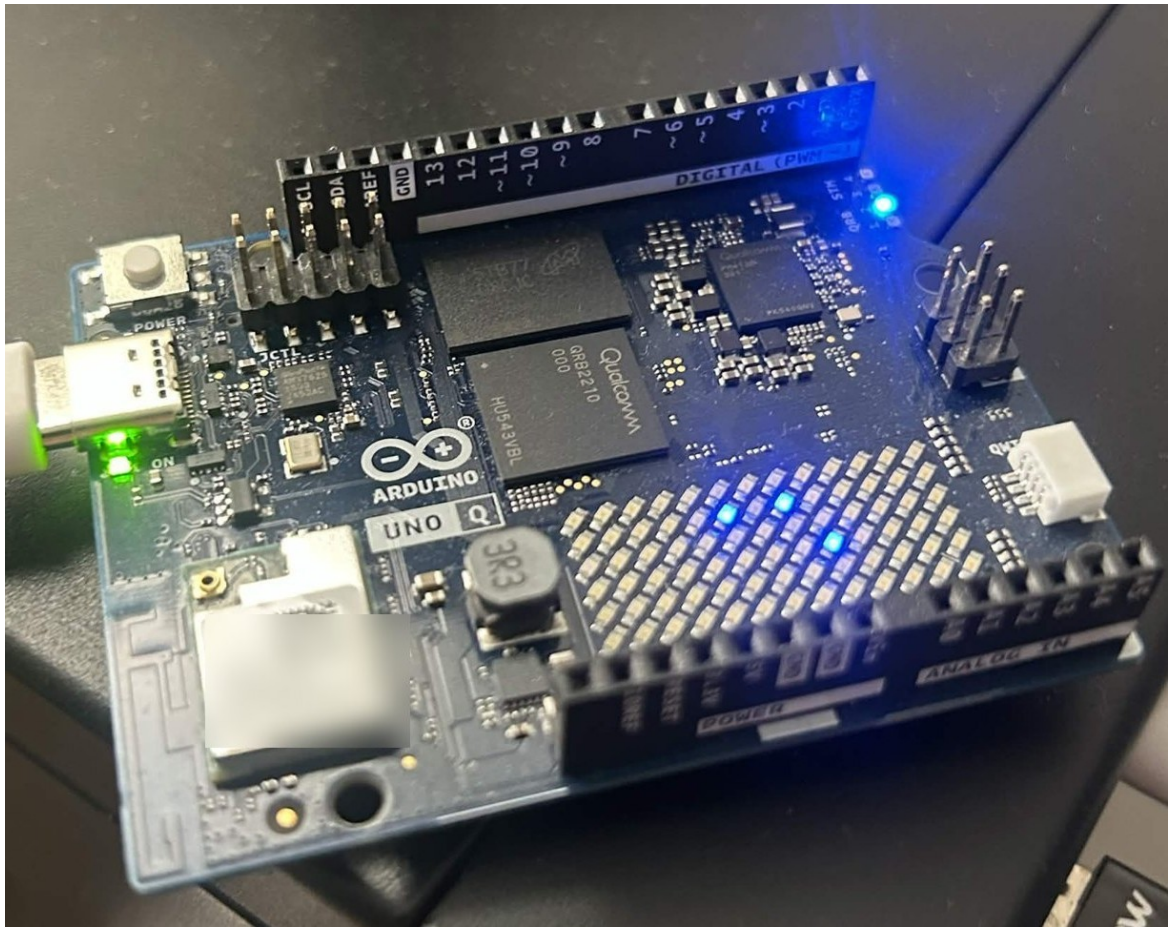
USB-C
power; draw not measured

Offline
the app binds to the board and makes no network calls

About 90 ms
per app frame; the tissue check costs more than the model

Median stage times on the board at 4 threads. Web page time: second tissue pass plus JPEG. The model alone uses 34 MB of RAM.

Implementation on Arduino UNO Q



18.7 ms

per tile, model inference: int8, 4 threads, offline

93.05%

balanced accuracy on all 7,180 test tiles, below every teacher (94.13–95.55%)

6.0 MB

5.65 M parameters: 1.4% of the two teachers' vision towers

All measured on the Arduino UNO Q: Qualcomm QRB2210, 4× Cortex-A53, 3.6 GiB RAM, onnxruntime on CPU. Stored test tiles; no camera captures yet.

How we built it

1 → 2 → 3 → 4 → 5

Plan
Design, plan and data splits written down before the code

Read
CONCH and MUSK on one RTX 5060: 20k of 100k training tiles

Freeze
Rules chosen on validation and pre-registered; test scored once

Train
Our student distilled from both, with K2 camera effects, then int8

Deploy
Flashed to the UNO Q; everything re-measured on the board

What broke	What we did
Two int8 calibration methods collapsed: only 44% of calls survived	→ **Percentile calibration: 98.9% survive on validation**
int8 scored lower on x86 (92.19%) than on the board (93.05%)	→ **Traced to one onnxruntime kernel, by running on real hardware**
CONCH and MUSK licences bar sharing their student	→ **Open control pair, with a shareable bundle**

Future Directions

What remains	Next step
The fusion gain is fragile: validation chose the rule by 1–2 tiles	→ **Rerun on all 100k training tiles on NCSA Delta (kit ready)**
The pre-registered threshold flagged 25.1% of tiles, not 10%	→ **Calibrate per site; add slide IDs for clustered CIs**
The board’s flag tracks the teachers weakly (κ 0.32), on stored tiles only	→ **Real camera captures on the board, then clinical validation**

Why it matters to pathologists and patients

For pathologists

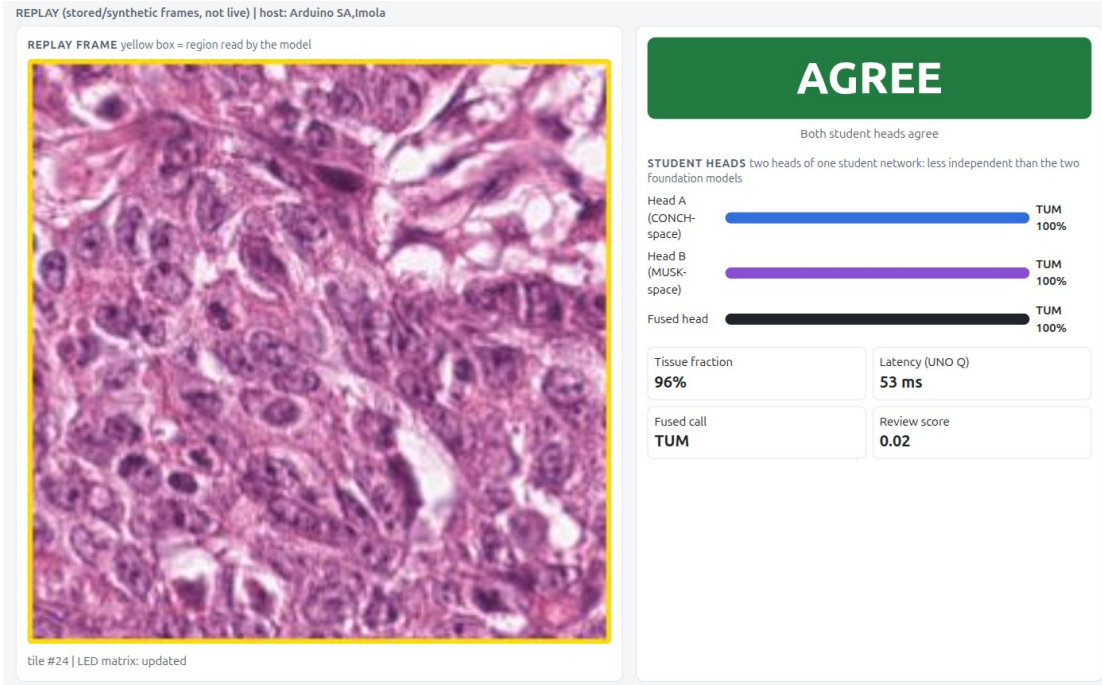
A short list of tiles worth re-checking, with each reader’s call shown.

For patients

A low-cost offline second look could help where no second pathologist is available.

For the field

Two readers help by making a better-calibrated reader, not through disagreement.



The app on the UNO Q, replaying a stored test tile. Its two heads share one student network.

TwinRead in three numbers

+0.74 pts

balanced accuracy from fusion on colorectal tiles

80.3%

of errors caught by reviewing the least certain 10% of tiles

18.7 ms

per tile on an offline board: a 6.0 MB model, 93.05% accurate

Thank you. Questions?

Yubo Wu, Noah Zhang, William Liu, Franklin Liu, Andy Chen

github.com/iujab/twinread-conch-musk-edge

References

Elmore JG, et al. Diagnostic concordance among pathologists interpreting breast biopsy specimens. JAMA 2015. doi:10.1001/jama.2015.1405

Fleming KA, et al. The Lancet Commission on diagnostics: transforming access to diagnostics. Lancet 2021. doi:10.1016/S0140-6736(21)00673-5

Wilson ML, et al. Access to pathology and laboratory medicine services: a crucial gap. Lancet 2018. doi:10.1016/S0140-6736(18)30459-8

Lu MY, et al. A visual-language foundation model for computational pathology (CONCH). Nat Med 2024. doi:10.1038/s41591-024-02856-4

Xiang J, et al. A vision–language foundation model for precision oncology (MUSK). Nature 2025. doi:10.1038/s41586-024-08378-w

Ikezogwo WO, et al. Quilt-1M: one million image-text pairs for histopathology (QuiltNet). NeurIPS 2023. arXiv:2306.11207

Li et al. Multi-modal foundation models for computational pathology: a survey. TMLR 2025. PMC13081565

Muneer A, et al. Review on multimodal data integration for cancer research. Artif Intell Rev 2026. doi:10.1007/s10462-026-11522-9

Kather JN, et al. NCT-CRC-HE-100K and CRC-VAL-HE-7K (CC-BY-4.0). Bandi P, et al. Camelyon17; Koh PW, et al. WILDS (Camelyon17-WILDS, CC0).

Open models: PathGen-CLIP (CC-BY-4.0), QuiltNet-B-16 (MIT), timm mobilenetv4_conv_small.

FCC. Communications Status Report for Areas Impacted by Hurricane Maria. 23 September 2017.

McBrayer KT, Chai PR, Judd EL. Communication Delays, Disruptions, and Blackouts for Crewed Mars Missions. NASA, ASCEND 2022. NTRS 20220013418.

NASA Science. Perseverance’s SuperCam uses AEGIS for the first time. 2022.

Arduino. A heads-up on the Arduino UNO Q board pricing. Arduino blog, 26 June 2026.

Code, results JSON and this deck’s generator: github.com/iujab/twinread-conch-musk-edge. Every TwinRead number here is read from results/*.json at build time.

Backup: data, splits and statistics

Split	Data	Tiles	Used for
Train	NCT-CRC-HE-100K subset	17,349	Probes and student
Validation	Same source, group-disjoint	2,658	Fusion rule, thresholds
Test	CRC-VAL-HE-7K, 9 classes	7,180	Evaluated once, after pre-registration
Test	Camelyon17-WILDS, hospital 2	1,500	Unseen hospital; trained on hospitals 0, 3, 4

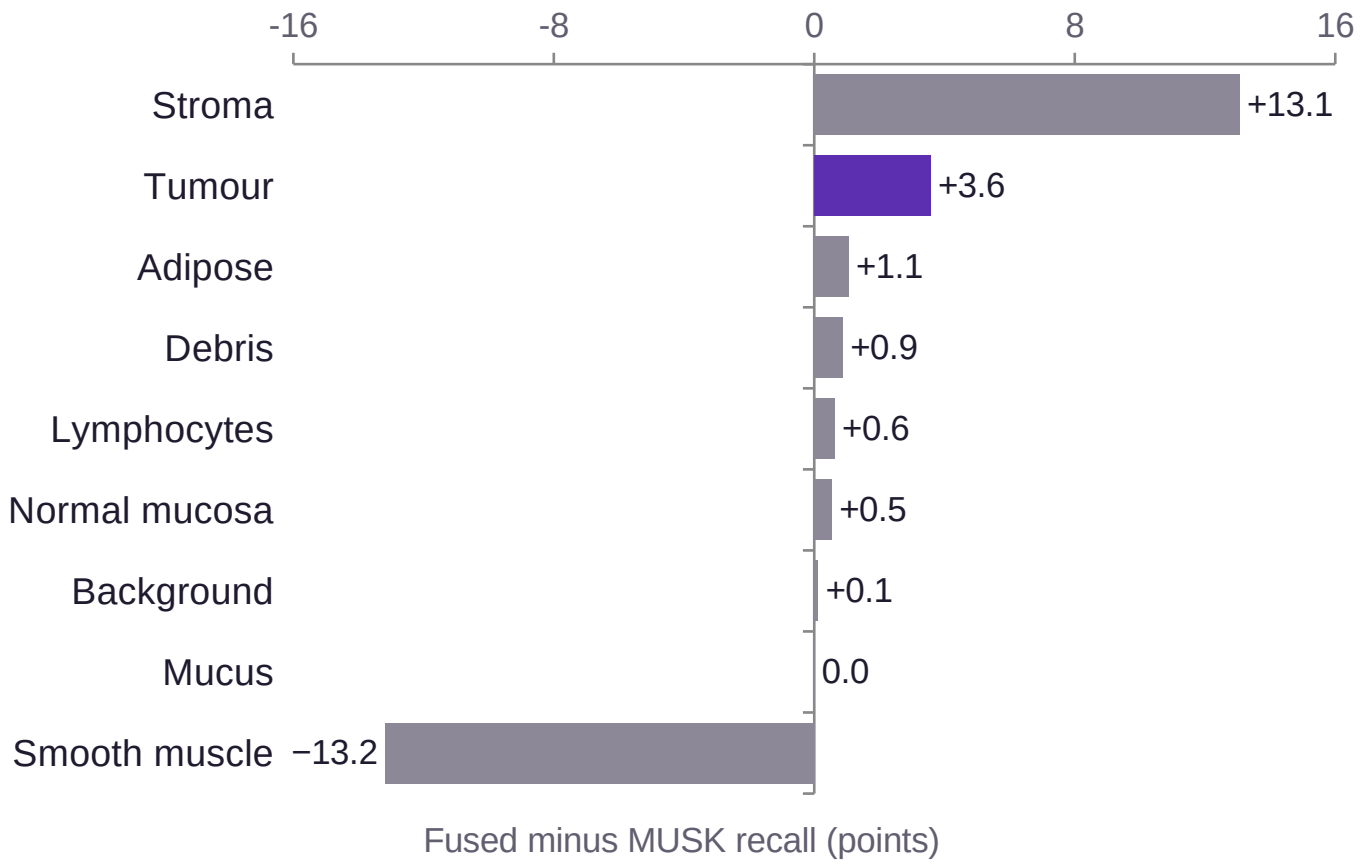
Statistics. Percentile bootstrap, 1,000 resamples. Colorectal CIs and McNemar tests are tile-level and anti-conservative because the dataset has no slide IDs; Camelyon17 CIs are clustered by slide.

Licences. CONCH and MUSK are CC-BY-NC-ND, so their student is never distributed. PathGen-CLIP + QuiltNet-B-16 is the control pair and the shareable bundle.

Scale. Teacher numbers are from a local 20k-tile subset on one RTX 5060. The Delta kit is written and tested locally and has not run on Delta.

Sources: results/cm_local/CM/*/preregistration.json, results/student/cm_crc/metrics.json, ncsa/README.md.

Backup: why the colorectal gain is fragile



+0.40 of the +0.74

points of balanced accuracy come from tumour recall. Muscle and stroma cancel.

2,641 / 2,640 / 2,639

of 2,658 validation tiles correct for the average, gated MLP and concat rules. Validation is near ceiling.

−0.40

test points for the runner-up rule, the gated MLP, against MUSK.

Post hoc and descriptive. Balanced accuracy is the mean of 9 class recalls, so each class contributes its recall change divided by 9. Source: results/headline.json.

Backup: where the review flag falls short

At a 10% budget	Fused uncertainty	Learned flag	Best single reader	Flag: cleared-tile accuracy	Best single: cleared-tile accuracy
Colorectal	80.3%	80.3%	74.1%	99.27%	98.85%
Camelyon17	81.1%	75.5%	74.5%	99.04%	99.11%

Pre-registered threshold on test	Target flagged	Actually flagged	Errors caught
Colorectal	10%	25.1%	97.5%
Camelyon17	10%	5.3%	58.5%

Validation sits near ceiling (99.36%), and validation to test is a distribution shift, so a threshold calibrated on it does not transfer. Recalibrate on the target site.

Single readers flag their own errors by their own confidence. Source: results/cm_local/CM/*/review.json, results/headline.json.

Backup: kernel throughput

K1: tiles/s, both readers, 224 px	Rate
PIL, 1 CPU process	221
PIL DataLoader, 8 workers	538
torch GPU ops (exact)	1,340
Triton (our baseline)	82,058
K1 CUDA, including host copy	46,157
K1 CUDA, input on GPU	**90,131**

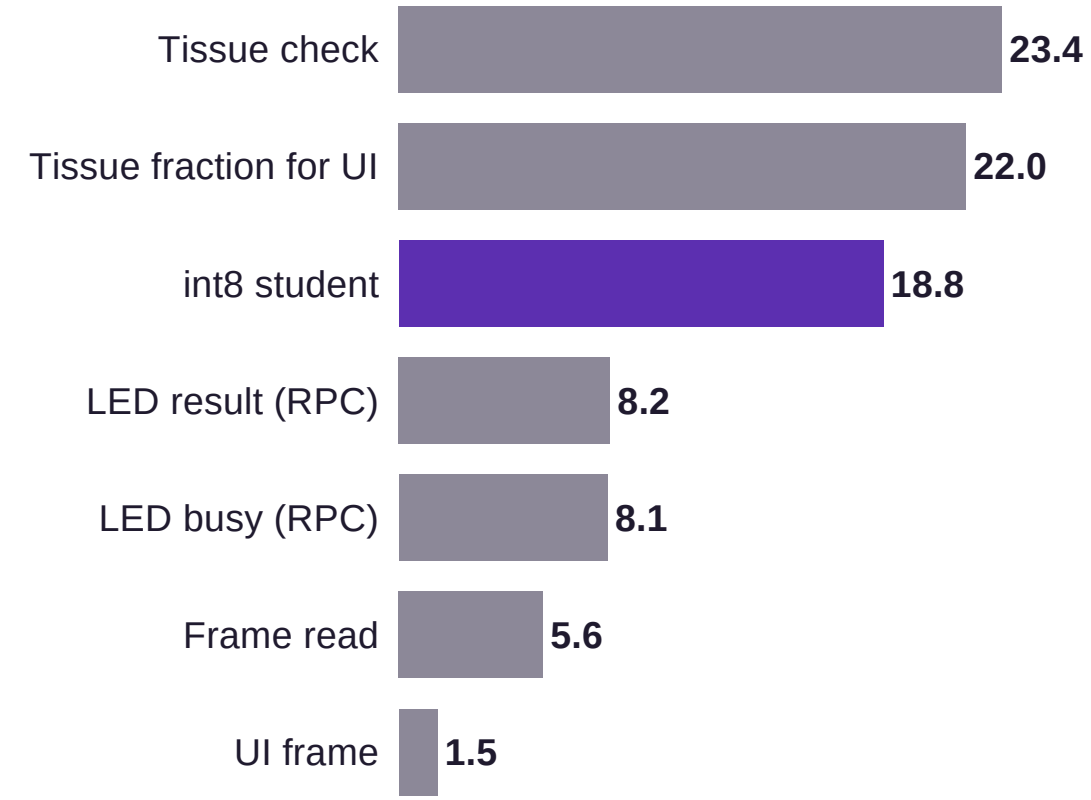
K2: images/s, severity 1.0	Rate
numpy, 1 process	44
numpy, 4 processes	166
torch GPU ops	1,097
K2 CUDA	**46,315**

K1 cosine to the official transforms: at least 0.999997. The ×1.11 end-to-end figure is for a batched, GPU-forward-bound run, where preprocessing is 9.5% of the time. Per-image serving and faster GPUs are a different regime, not measured here.

CUDA-event timing (warmup 3, median of 10) on a shared GPU, so throughput is an estimate. Source: results/kernels/*.json.

Backup: a 90 ms app frame and int8 fidelity

Median time per stage (ms)



CONCH + MUSK student, int8	Balanced accuracy
On the board	**93.05%**
x86, fused int8 kernels off	93.01%
x86, onnxruntime defaults	92.19%
fp32 student	94.15%

Board vs x86 kernels off: 13 of 7,180 calls differ.

Board flag: 10.2% of tiles, catches 57.2% of its own errors, agreement with the teacher flag 79.6% (κ 0.32).

The device's two readers are two heads on one shared backbone.

Full app loop 25.4 tiles/s. Each Bridge RPC call to the MCU costs about 8 ms. Source: results/uno_q/onboard/*.json.